Does the NBA Pay for Production? Last Season's Efficiency and Player Salaries, 2026-27

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Repository: [https://github.com/jonathan-ha1/Econ_3916_Midterm.git]

Notebook:
[https://colab.research.google.com/drive/1WgTnEe_VhA7fbsRdGCxyFwMG8wZVwSqX?usp=sharing]

Question	Among NBA players in the 2026-27 season, how is a player's efficiency last season (2025-26 Box Plus/Minus, BPM) related to his 2026-27 salary (US dollars, in logs), controlling for age and position?
Hypothesis	I expect a positive coefficient on BPM, since teams set contracts based on how players have already performed, so players who were more efficient last season should be getting paid more now.
Headline finding	Holding age and position fixed, one more point of BPM last season is associated with about 17.0% higher salary this season, with a 95% confidence interval of 13.2% to 21.0% and N = 345. This is a conditional association and not a causal effect, mainly because a lot of salaries were set by contracts signed before last season even started.
Data	Basketball-Reference's 2026-27 player contracts and 2025-26 advanced stats. One row is one player, and the final sample is 345 players who played at least 500 minutes in 2025-26.
Method	A regression (OLS) that predicts each player's log salary from his BPM last season, his age, and his position, with centers as the group the other positions are compared to. I use regular standard errors, since a test (Breusch-Pagan) showed the model's errors stay about the same size across players, and robust standard errors give almost the same answer.

1. The Question and the Data

Salary in the NBA is often treated as a reward for how well a player plays, but this leaves out how much of a player's pay is set by things other than performance. Salary is better understood as the result of a contract system, where league rules, timing, and age shape pay alongside production. Based on the data, last season's efficiency is strongly related to this season's salary, but that relationship should be read as an association and not as the price teams pay for performance.

This question matters because every team must fit its roster under a salary cap, so money spent on one player cannot be spent on another. Front offices need to know whether they are paying for production or for something else, and players and agents need to know what kind of performance gets rewarded when they negotiate a contract. If salary follows efficiency closely, this means the market is paying for how

players perform, while a weak relationship would suggest that reputation, age, or contract timing matter more than how someone played.

Both datasets come from Basketball-Reference, a site run by Sports Reference that collects NBA stats and publicly reported contract information for fans, media, and analysts. The salary data show what each player is set to make in 2026-27 under his contract, not money he has already been paid, which means the numbers reflect what was agreed to when he signed. The stats come from the site's 2025-26 advanced table, where Box Plus/Minus (BPM) estimates how many points per 100 possessions a player adds compared with a league-average player. BPM is not something that can be directly observed, since it is built from box score stats and adjusted to team results, which makes it a measurement choice of its own. This matters because BPM picks up offense better than defense and gets noisy for players who barely played. To put the data together, I removed duplicate salary rows for players a former team still owes money to, since their 2026-27 salary was the same in both rows, kept the combined row for players traded during 2025-26, and merged the two files on Basketball-Reference's player ID so that accents and suffixes in names would not cause mismatches.

After merging, none of the variables I use have missing values, but two groups of players are left out before that point. In total, 58 players with a 2026-27 salary have no 2025-26 stats, mostly rookies and players who didn't play last season, and I also removed 52 players who played under 500 minutes, since BPM isn't reliable for players who barely played. This is MNAR, because players usually miss a season or barely play when they're injured or not playing well enough to earn minutes, and both of those reasons are tied to the BPM they would have had. Because of this, my final sample of 345 leans toward healthy, established players, so the results show how the league pays players who are already in a team's rotation, not every player under contract.

Table 1 shows each step from the full salary list to the final sample, and Table 2 summarizes the 345 players who remain.

Table 1. From players with a 2026-27 salary to the sample

Step	Players
Players with a 2026-27 salary	455
No 2025-26 statistics (rookies, did not play)	−58
Under 500 minutes in 2025-26	−52
Players in the sample	345

Steps are applied in order, so a player is counted once, at the first step he fails. Duplicate salary rows for players still paid by a former team are removed before the first step.

Table 2. Summary statistics, 345 players, 2026-27 salaries and 2025-26 statistics

Variable (units)	Mean	Median	SD	Min	Max	N
Salary, 2026-27 (US dollars)	15,134,022	9,160,715	15,099,477	2,055,000	62,587,158	345
Log salary, 2026-27	16.05	16.03	1.00	14.54	17.95	345

Variable (units)	Mean	Median	SD	Min	Max	N
BPM, 2025-26 (points per 100 possessions)	-0.08	-0.40	2.91	-7.30	14.20	345
Age (years)	25.98	25.00	4.23	19.00	41.00	345
Minutes played, 2025-26	1,520.12	1,455.00	572.94	500.00	2,953.00	345

Note: Salary is from Basketball-Reference's player contracts page for the 2026-27 season, and BPM, age, and minutes are from Basketball-Reference's 2025-26 advanced statistics. Position (PG, SG, SF, PF, C) comes from the same statistics file and enters the model as dummy variables.

Table 2 shows that salary is heavily right-skewed, which is why the model uses log salary instead of regular salary. The average salary is 15.1 million dollars, but the median is only 9.2 million dollars, since a few stars make over 50 million dollars, with the highest salary in the sample at 62.6 million dollars, while a lot of players are close to the minimum. Raw salary has a skew of 1.39, and using the log brings it down to 0.17. Average BPM is about 0, which makes sense since it's measured against a league-average player, and it ranges from -7.3 to 14.2, while the typical player is 25 years old and played about 1,450 minutes.

2. Descriptive Evidence

The data shows that salary is heavily skewed, and players with a higher BPM tend to get paid more, though with a lot of spread at every level of BPM.

NBA salaries are set inside a system of league rules, with a minimum salary at the bottom, maximum contracts at the top, and rookie-scale deals for players early in their careers, so salary is far from evenly spread out. The mean salary of 15.1 million dollars sits far above the median of 9.2 million dollars, which shows that the average is pulled up by a small group of very highly paid players, while about half of players earn under 9.2 million dollars. Raw salary has a skew of 1.39, which is above the point of about 1 where a log is worth using, and the log brings it down to 0.17, so I use log salary as the dependent variable throughout. This matters for more than just the shape of the data, since coefficients in a log model can be read as rough percent changes in salary, which is how teams and agents usually talk about raises and pay gaps.

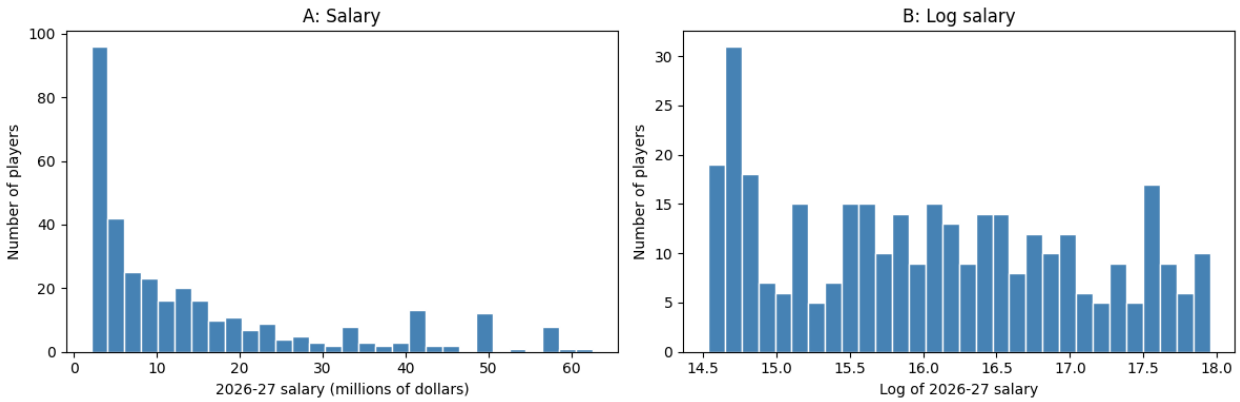


Figure 1: Salary is very uneven, since about half of players make under 9.2 million dollars while the highest-paid player makes 62.6 million dollars, which gives raw salary a skew of 1.39 (Panel A). Using the log brings the skew down to 0.17 and spreads players out more evenly (Panel B), which is why the model uses log salary, but there's still a spike on the far left from players making the league minimum, since their pay is set by league rules and not by how well they played.

Figure 1 also shows something the log does not fix, which is a spike of players at or near the league minimum. These players cannot be paid less no matter how they played, so their salary is set by league rules instead of by their performance, and this same floor shows up again in the residuals in Section 4.

Figure 2 plots each player's 2025-26 BPM against his 2026-27 log salary, and the relationship is clearly positive, with a correlation of 0.50 between the two. Almost every player with a BPM above 5 earns far more than the league average, which fits the idea that teams reward clear top performers. Below that level, though, players with similar BPMs earn anything from the minimum to tens of millions of dollars, and a flat band along the bottom shows minimum-salary players spread across a wide range of BPM.

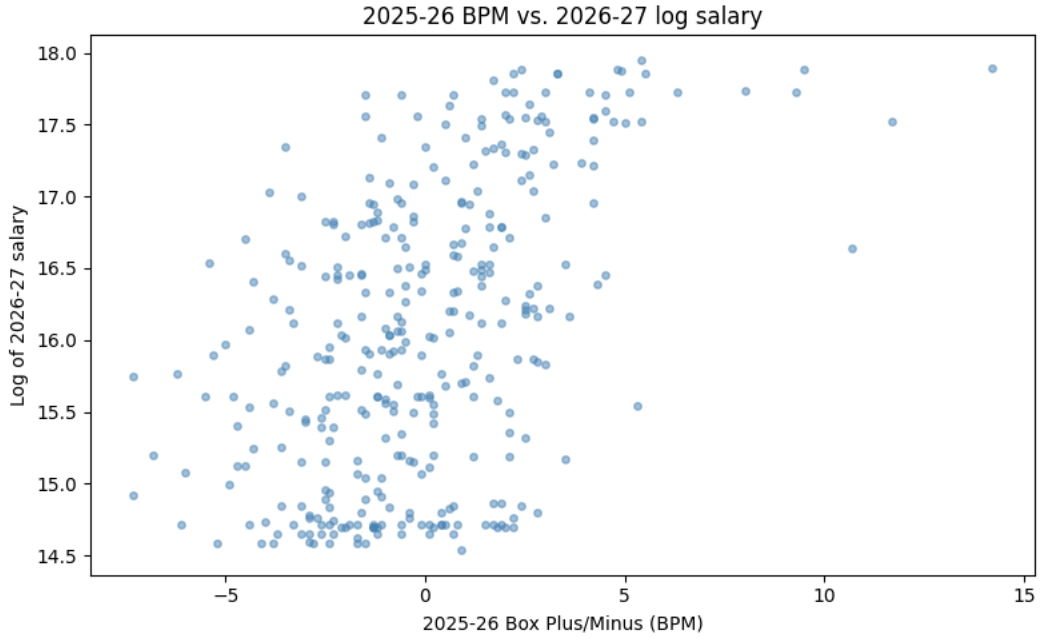


Figure 2. Players with a higher BPM last season tend to earn more this season (correlation of 0.50), but BPM is far from the whole story, since players with similar BPMs earn very different salaries and a band of minimum-salary players spans a wide range of BPM. Source: Basketball-Reference, 2025-26 advanced statistics and 2026-27 player contracts.

The wide spread in salary at each level of BPM is what makes controls necessary in the Section 3 model. Age affects both how players perform and how they get paid, as younger players are usually on fixed rookie-scale contracts while veterans sign bigger contracts later in their careers, and position can affect both a player's BPM and how teams value his skills, so the model accounts for both instead of relying on BPM alone. Both figures have axes labeled by unit and season, histogram counts starting at zero, no data cut off, and no unnecessary decoration. The scatter plot's vertical axis does not start at zero, which is normal for scatter plots, because a log salary of zero is far below any salary in the data and only bar charts need to start at zero.

3. The Model

The central question of this section is how much more players are paid when they performed better in the previous season, and after accounting for age and position, each additional point of BPM is associated with about 17.0% higher salary. This estimate comes from an OLS regression across all 345 players:

Log salary = constant + BPM + age + age squared + position + error term

The model predicts each player's salary based on their performance in the previous season, their age, and their position, while the error term represents all factors the model does not capture, such as the timing of a player's contract or their reputation around the league. Centers serve as the comparison group, so the model evaluates whether players at other positions are paid more or less than centers of the same age and performance level. A second version of the model adds minutes played, and both versions are reported in Table A1 in the appendix.

The main result shows that stronger performance is associated with higher pay, as a player with one additional point of BPM in the previous season earns about 17.0% more this season than a player of the same age and position. This estimate is also precise, since the 95% confidence interval ranges from 13.2% to 21.0%, indicating that even the most conservative estimate reflects a meaningful increase in salary. In practical terms, a player who performed slightly above the league average would be expected to earn about 17% more than an otherwise similar player at the average.

Age also plays a significant role, although its relationship with salary is not linear, as pay increases early in a player's career and then declines, reaching its peak at around age 31. This pattern is consistent with the structure of NBA contracts, since younger players are bound by league-set rookie contracts, players typically sign their largest contracts in their late twenties and early thirties, and older players tend to accept smaller contracts as their performance declines.

Each control is included for a specific reason, and age is included because it affects both a player's performance and their salary, so excluding it would distort the estimated effect of BPM. Age cannot serve as a bad control, since performance cannot change a player's age, while age squared allows the model to capture the rise-and-fall pattern rather than forcing a straight-line relationship. Position is included for a similar reason, as larger players such as centers tend to record higher BPMs through rebounds and blocks, and teams may value positions differently. Position would only be a bad control if strong performance caused players to change positions, but position is primarily determined by a player's size and role rather than their efficiency in the previous season. The position estimates are not precise enough to show clear differences in pay between positions, but their inclusion prevents those differences from affecting the BPM estimate.

Minutes played is excluded from the main model, since more efficient players tend to receive more playing time, which makes minutes one of the channels through which BPM leads to higher pay, and including it would therefore remove part of the effect being measured. When minutes is added in the second version of the model, the effect of BPM falls from 17.0% to 12.5%, indicating that part of the reason better players earn more is that they also play more.

Overall, the model explains about 30% of the variation in salary across players, which indicates that most of the differences in pay come from factors outside the model, such as when a player signed their contract and their reputation in the league, and these limitations are discussed further in Section 5.

4. Diagnostics

The diagnostics support the main model, since the residuals don't show any curve or fan shape, the constant-variance test passes, the VIFs don't flag anything for BPM, and no small group of players is throwing off the result.

The Model 1 residuals are plotted against the fitted values and against BPM (Appendix, Figure A1), and in both plots the points are spread evenly above and below zero. There's no curve that would mean the model is missing something and no fan shape that would mean the errors get bigger as salary or BPM goes up, so a straight-line model works here. The one clear pattern is a straight diagonal line along the bottom of the residuals vs. fitted plot, which comes from players making the league minimum, since their salary can't go any lower, so the more the model expects them to make, the further their residual drops along that line. This is the same salary floor from Figure 1, and it means the model expects some players to be paid more than they are, when their salary is actually set by league rules and not by how they played.

OLS assumes the errors are about the same size for every player, and the Breusch-Pagan test checks this, coming out with a p-value of 0.3545, which is far above 0.05, so there's no sign of a problem and the regular standard errors from Section 3 are fine to use. As an extra check, the model was also run with HC3 robust standard errors, which don't need that assumption, and they came out almost the same, with BPM's standard error at 0.0169 regular and 0.0152 robust, so the results hold up either way.

VIFs check whether any variables in the model are too similar to each other, and BPM's VIF is 1.16 while the position dummies are between 1.66 and 1.90, which are all low, so these variables aren't too similar to anything else in the model. Age and age squared both have VIFs around 100, which would usually be a problem, but it's expected here since age squared is made from age, so the two go up and down together. That makes the age results a little less exact on their own, but it doesn't change the BPM result at all, and together the two age terms still show salary peaking at age 31.0.

To see if a few players were driving the whole result, Cook's distance was calculated for every player, and the five biggest were LeBron James, DeMar DeRozan, Mike Conley, Nikola Vučević, and Stephen Curry. Four of them are older players on cheap contracts, like LeBron, who is 41 and making about 3.9 million dollars even with a BPM of 3.5, and they stand out because they're at the edge of the age range and the model expects good veterans to get paid more, when a lot of players near the end of their careers take smaller deals. When Model 1 is run again without these five players, the BPM coefficient barely changes, going from 0.1574 to 0.1561, so the main result isn't being carried by a handful of unusual players.

5. Threats and Conclusion

The BPM coefficient is often read as the price teams pay for performance, but this leaves out how contracts actually get set. For the 17.0% figure to be causal, meaning that raising a player's BPM by one point would raise his salary by 17.0%, BPM would have to be unrelated to every other part of salary once age and position are held fixed, and that is unlikely here.

The biggest threat is contract timing, since a lot of 2026-27 salaries come from contracts signed two or more years earlier, which means they were set based on how a player performed before signing and not on his 2025-26 BPM. For those players, last season's BPM is only a rough stand-in for the performance teams actually paid for, so players whose BPM changed since signing will look overpaid or underpaid in this data. There are also things the model cannot see that likely affect both performance and pay, such as

reputation, marketability, injury history, and draft position. A player like Stephen Curry earns a top salary partly because of his long record and standing in the league, not just because of last season's BPM, so part of the BPM coefficient may be picking up those factors instead.

Rookie-scale contracts add another layer, since salaries for players early in their careers are set by the league's pay scale rather than negotiated based on performance. Age partly accounts for this, but the data does not identify which players are on rookie deals, so their fixed salaries may weaken the measured link between BPM and pay. The sample also leaves out rookies, injured players, and players with under 500 minutes, a pattern of missingness labeled MNAR in Section 1, so the result describes established rotation players and may not hold for the full league. BPM itself is an estimate that captures defense less well than offense, which means some of what teams pay for is not reflected in it at all. This question also involves no policy change with a before-and-after comparison group, so a difference-in-differences design does not apply. Because of these threats, the estimate is best described as a conditional association: among established players with at least 500 minutes, a player whose BPM was one point higher last season earns about 17.0% more this season than a player of the same age and position.

Overall, how well a player performed last season is strongly related to how much he earns this season, with each additional point of BPM linked to about 17.0% higher pay, holding age and position fixed. A reader can be fairly confident this association is real and not just chance or a few unusual players, since the 95% confidence interval runs from 13.2% to 21.0%, the Breusch-Pagan test found no sign that the model's errors get bigger or smaller across players, and removing the five most influential players barely changes the estimate. A reader should not treat it as the exact price teams pay for one point of BPM, though, since the association falls to 12.5% once minutes are held fixed and contract timing and reputation likely shape it as well. With another month, I would match each player's salary to his stats from the season his contract was signed, add more seasons of data with season dummies so the estimate does not rest on one year, and identify rookie-scale contracts so they could be looked at separately from contracts negotiated on the open market.

As a whole, the NBA does pay for production, but not in a simple way, since last season's efficiency explains a meaningful part of this season's salary while contract timing, age, and league rules shape the rest.

References

Sports Reference LLC. "2025-26 NBA Player Stats: Advanced." Basketball-Reference. https://www.basketball-reference.com/leagues/NBA_2026_advanced.html. Retrieved 2026-11-04.

Sports Reference LLC. "NBA Player Contracts." Basketball-Reference. https://www.basketball-reference.com/contracts/players.html. Retrieved 2026-11-04. 2026-27 salary column used.

Appendix

Table A1. Log 2026-27 salary regressed on 2025-26 BPM and controls

	(1) Main	(2) Adds minutes
BPM, 2025-26	0.1574	0.1177
	(0.0169)	(0.0177)

	(1) Main	(2) Adds minutes
Age (years)	0.3485	0.3671
	(0.1084)	(0.1039)
Age squared	-0.0056	-0.0059
	(0.0020)	(0.0019)
Point guard (0/1)	0.2003	0.1109
	(0.1547)	(0.1491)
Shooting guard (0/1)	0.2220	0.1280
	(0.1447)	(0.1396)
Small forward (0/1)	0.2771	0.1548
	(0.1526)	(0.1479)
Power forward (0/1)	0.3568	0.3228
	(0.1491)	(0.1430)
Minutes, 2025-26		0.0005
		(0.0001)
Constant	10.6900	9.7518
	(1.4842)	(1.4318)
N	345	345
R^2	0.303	0.362

Note: Dependent variable: log of 2026-27 salary in US dollars. OLS with classical standard errors in parentheses. Centers are the reference group for the position dummies. Column 1 is the main model. Column 2 adds 2025-26 minutes played, which lies on the path from BPM to salary (Section 3), so it is a robustness check rather than the main specification.

Table A2. Classical vs. HC3 robust standard errors (Model 1)

Variable	Coefficient	Classical SE	HC3 SE
Constant	10.6900	1.4842	1.5481
BPM	0.1574	0.0169	0.0152
Age	0.3485	0.1084	0.1166
Age squared	-0.0056	0.0020	0.0022
Point guard	0.2003	0.1547	0.1604

Variable	Coefficient	Classical SE	HC3 SE
Shooting guard	0.2220	0.1447	0.1481
Small forward	0.2771	0.1526	0.1479
Power forward	0.3568	0.1491	0.1512

Note: Breusch-Pagan test p-value = 0.3545.

Table A3. Variance inflation factors (Model 1)

Variable	VIF
BPM	1.16
Age	100.73
Age squared	99.91
Point guard	1.66
Shooting guard	1.90
Small forward	1.71
Power forward	1.75

Table A4. Five most influential players by Cook's distance (Model 1)

Player	Age	BPM	Salary, 2026-27 (US dollars)	Cook's distance
LeBron James	41	3.5	3,876,529	0.0705
DeMar DeRozan	36	-0.1	2,449,421	0.0216
Mike Conley	38	-1.3	2,449,421	0.0213
Nikola Vučević	35	1.5	2,449,421	0.0158
Stephen Curry	37	5.4	62,587,158	0.0146

Note: BPM coefficient with all 345 players = 0.1574; without these five = 0.1561.

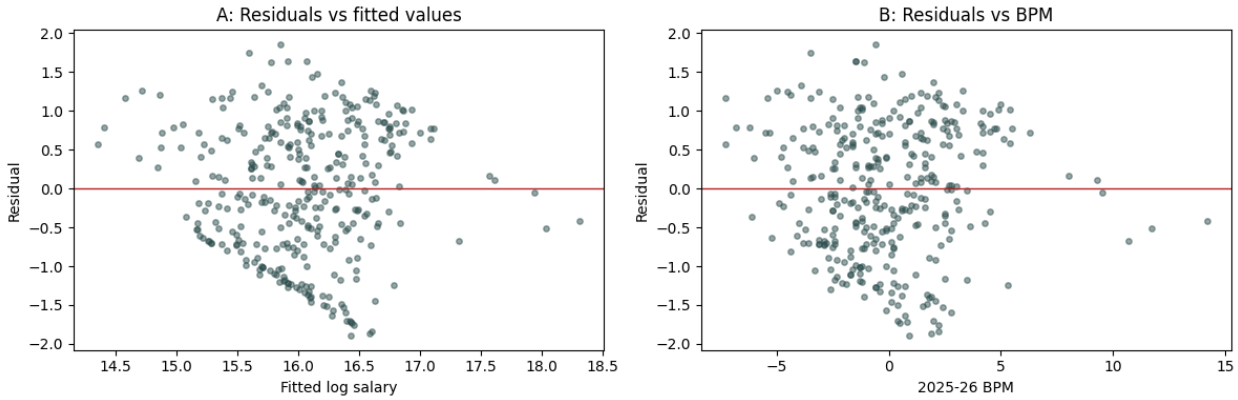


Figure A1. Residuals show no curve or fan shape against fitted values or BPM; the straight diagonal edge at the bottom of Panel A comes from players on the league minimum salary.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the download and cleaning steps and the order to run the cells.